	$\mathscr{B}_1$	$\mathscr{B}_2$	$\mathscr{B}_3$	$\mathscr{B}_4$
$\mathscr{B}_1$	0.367	0.625	0.944	0.833
$\mathscr{B}_2$	0.708	0.750	0.528	0.375
$\mathscr{B}_3$	0.056	0.167	0.194	0.722
$\mathscr{B}_4$	0.250	0.250	0.667	1.000

Fig. 9.11. Density table for the advice relation from Krackhardt's high-tech managers, positions identified by hierarchical clustering of correlations

	$\mathscr{B}_1$	$\mathscr{B}_2$	$\mathscr{B}_3$	$\mathscr{B}_4$
$\mathscr{B}_1$	0	1	1	1
$\mathscr{B}_2$	1	1	1	0
$\mathscr{B}_3$	0	0	0	1
$\mathscr{B}_4$	0	0	1	1

Fig. 9.12. Image matrix for the advice relation from Krackhardt's high-tech managers, positions identified by hierarchical clustering of correlations

of $\mathscr{B}_1$ to $\mathscr{B}_2$. In Figure 9.10 we see that 15 of these 24 possible choices are present, so the density of this submatrix is equal to 0.625. This is the entry in row 1, column 2 of the density table. Figure 9.11 shows the density table for this example. These densities were calculated using *UCINET 3* (MacEvoy and Freeman n.d.).

Image Matrices. Often we would like to summarize the ties between positions in a more parsimonious way. An *image matrix* is a summary of the ties between and within positions, so that each tie is coded as either present or absent between each pair of positions. If submatrices are filled completely with 1's (oneblocks) or completely with 0's (zeroblocks) then the decision concerning whether a tie exists between positions is straightforward. However, since actual network data are seldom so perfect, we need a guideline for deciding whether a tie exists between positions.

There are several rules for constructing an image matrix from a density table, and we discuss these in detail in Chapter 10. For now, let us illustrate with one rule, the α density rule. This rule specifies a tie as present between two positions if the density of ties from actors in one position to actors in another position is greater than or equal to the density of the matrix as a whole. Letting Δ be the density of ties for the relation, we define a tie as present from position $\mathscr{B}_k$ to position $\mathscr{B}_l$ if the

density of ties from members of po
is greater than or equal to Δ.

Using this rule on the four-positi
Krackhardt's high-tech managers f
correlations gives the image matrix
entire sociomatrix is 0.452. Cons
position $\mathscr{B}_1$ to members of positio
actors in position $\mathscr{B}_1$ to actors in
the density for the whole sociomatr
to $\mathscr{B}_3$ in the image matrix. Howe
present from members of position
to 0.056. Since this is less than the
a tie as absent from position $\mathscr{B}_3$ to
between densities coded as present
those coded as absent (the *largest* o

Image matrices are fundamental
all of Chapter 10 to blockmodels, w
topic until then.

Reduced Graphs. A third us
and within positions is in a *reduced*
are represented as nodes and ties be
define the arcs between nodes. It is
from the image matrix. A "1" in th
is an arc from the node represent
representing the column position in
graph, there is an arc from the no
node representing position $\mathscr{B}_l$ if ther
matrix.

Figure 9.13 shows the reduced grap
of the advice relation for Krackhard
four positions is represented by a
represent ties present in the image
the president and one of the vice pre
presidents) primarily receive ties, wh

9.6 Summary

In conclusion, let us consider some
especially positional analysis based